

Riya Gupta



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EDUCATION

Vishwakarma Institute Of Technology

Computer science (Artificial Intelligence) — CGPA: 9.25

Pune, IN

Sept. 2024 – April. 2028

Decent Junior College of Engineering

HSC Percentage: 75.5

Nagpur, IN

Sept. 2022 – May. 2024

Providence girls High School

1-10 Percentage: 93.28

Nagpur, IN

Sept. 2012 – May. 2022

PROJECTS

AI-Powered Flood Surveillance System | Python, YOLOv12, OpenCV, FastAPI, React, Leaflet.js, SQLite | [GitHub](#)

- Developed an AI-powered flood monitoring platform that analyzes drone/CCTV video clips to detect trapped individuals and provide real-time intelligence for emergency response.
- Implemented computer vision pipelines using **YOLOv12**, **OpenCV**, **MediaPipe**, **DeepFace** for people detection, demographic classification (adult/child/elderly), distress signal recognition, and flood depth estimation.
- Built a city-wide interactive disaster dashboard that visualizes flood severity, detected individuals, and rescue priorities across multiple monitored locations.

God's Hand (Rubik's Cube Solver) | C++, Pistache, RapidJSON

- Modeled a virtual Rubik's Cube using three different representations and developed the backend using the Pistache C++ framework.
- Implemented four solving algorithms including **BFS**, **DFS**, **IDDFS**, and **IDA***.
- Achieved a solving time of **under 1 second** for a Rubik's Cube scrambled up to 14 moves using the **Korf IDA* algorithm**.

WellCare – Smartwatch-based Health Monitoring System | Python, Arduino/ESP32, IoT, [GitHub](#)

- Designed and prototyped a wearable **IoT-based health monitoring smartwatch** capable of tracking vital parameters such as heart rate, SpO₂, and body temperature in real time.
- Integrated biomedical sensors with an **ESP32/Arduino embedded system** to continuously collect physiological signals and process health metrics.
- Implemented **Bluetooth-based wireless communication** to transmit health data in real time and generate emergency alerts for remote patient monitoring.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React.js, Next.js, Tailwind CSS, Node.js, Express.js, React Native, MongoDB, Git, Github

Developer Tools: VS Code, Linux, Git, GitHub, Docker, Postman, Neovim

Coursework: : Data Structures Algorithms, DBMS, Object-Oriented Programming, Competitive Programming, Operating System, Machine Learning